5. Domain: Artificial Intelligence

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	Next Generation Person Search
	Name of the Project under which technology is developed	Intelligent Video Surveillance and Video Analytics System
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	An analytical engine to perform rapid search of suspect/missing person of interest in surveillance videos. The person is searched given the description like time, direction of movement, gender, age and the colour and type of their dress. **Applications** Faster post-event investigation Enable missing person identification
	Details of Contact Person alongwith Mobile No. and e-mail ID	S Irene, C-DAC Chennai irenes@cdac.in 9940320384
2.	**Brief details of Technology Developed**	XRAYCAD- XRay Computer Aided Detection Tool for TB Analysis
	Name of the Project under which technology is developed	Design and Development of Chest X-Ray Image Processing Solutions for Computer Aided Diagnosis
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	The CHEST X-RAY Computer Aided Detection Software Solution by Centre for Development of Advanced Computing(C-DAC), Chennai is a Web Application Portal for assisting doctors in Chest X-Ray Diagnosis. This Solution is an Artificial Intelligence Engine that can help in Chest X-Ray Computer Aided Diagnosis. The functionality includes: a) Chest X-Ray Disease Classification b) Visualization of disease affected region c) Lung Zone Detection for Disease Manifestation d) Longitudinal Analysis. The solution is developed in collaboration with ICMR-NIRT
	Details of Contact Person alongwith Mobile No. and e-mail ID	Harikrishnan V S, C-DAC Chennai harikrishnans@cdac.in Mob:9790847089
3.	**Brief details of Technology Developed**	Ikshana-IOT device for Ocean Surveillance Solution
	Name of the Project under which technology is developed	Intelligent Video Surveillance and Video Analytics
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7

Sr. No.	Parameters	Brief details
	Brief Write-up on the Technology Developed	IoT Device Ikshana is an Artificial Intelligence enabled Device, which can detect and identify Ocean objects like ships, boats etc in its preview using Computer Vision Technologies. It also generates summarized videos with objects detected. This in turn saves a lot of storage space with relevant objects in the ocean being only stored and can transmit object minimal information with GPS co-ordinates when deployed in ships. It was designed for Ocean monitoring in collaboration with NIOT
	Details of Contact Person alongwith Mobile No. and e-mail ID	Harikrishnan V S, C-DAC Chennai harikrishnans@cdac.in Mob:9790847089
4.	**Brief details of Technology Developed**	The Comprehensive Reliability and Robustness Assessment Toolbox addresses critical challenges in AI for defense applications, including adversarial attacks, data corruptions, and quality issues. It provides systematic evaluation of robustness, reliability, and uncertainty in AI models under varied conditions. By estimating uncertainty caused by incomplete data or inherent randomness, it enhances decision-making safety in high-risk domains. The toolbox fills a gap in the AI community by offering a formal methodology for assessing AI system reliability. This ensures safer and more dependable AI-driven defense applications.
	Name of the Project under which technology is developed	Comprehensive Reliability and Robustness Assessment Toolbox for AI Applications (CRAT)
	Funding Agency	Scientific Analysis Group (SAG), Defence Research & Development Organisation (DRDO)
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	The Comprehensive Reliability and Robustness Assessment Toolbox (CRAT) provides the way to formally analyze the robustness against adversarial attacks, reliability and uncertainty of AI models and in-turn the AI applications. Features: • Robustness Evaluation - Tests AI models against corruptions & adversarial attacks. • Data Quality Analysis – Out of distribution detection, drift and bias detection. • Uncertainty Quantification – Measures prediction uncertainty for better interpretability. • Formal Verification – Validates model behavior under adversarial conditions. HPC-Enabled - The integration of HPC capabilities to enhance the efficiency and scalability of robustness evaluation processes.
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Ramesh Naidu Laveti, C-DAC Bangalore **ph:** +91 98443 34898, Mobile: **+91 91641 73007** Email: **rameshl@cdac.in**
5.	**Brief details of Technology Developed**	Integrating key paradigm shifts in mental health research, the technology focuses on the critical developmental window from the antenatal period to young adulthood and employs repeated, multimodal assessments. It

Sr. No.	Parameters	Brief details
		emphasizes exposomal determinants—including post-pandemic influences—while leveraging advances in genomics and gut microbiome analysis. The platform incorporates digital phenotyping, remote and distance assessments, disruptive medical technologies, and satellite-based measurements to capture long-term environmental exposures.
	Name of the Project under which technology is developed	Pathways To Resilience and Mental-Health (PARAM)
	Funding Agency	Indian Council of Medical Research (ICMR)
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	Pathways To Resilience and Mental-Health (PARAM) aims to establish a multi-centre, multi-disciplinary, collaborative team-science platform, to enable the longitudinal study of the Neuro developmental origins of resilience/vulnerability to psychopathology (mental illness) using clinical and AI-based methods. Features • A comprehensive biorepository • Automated data collection pipelines for collecting data from hospitals supporting many data modalities such as clinical reports, medical images, demographic data etc. • Pipelines for data curation, anonymization, and quality control • Multi-centre, multi-disciplinary, collaborative data platform AI-based solutions for the detection of a select set of mental disorders and analysis
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Ramesh Naidu Laveti, C-DAC Bangalore **ph:** +91 98443 34898, Mobile: **+91 91641 73007** Email: **rameshl@cdac.in**
6.	**Brief details of Technology Developed**	The AI-Based Procurement Automation initiative aims to modernize and streamline the highly manual and time-intensive public procurement process currently used at NRL. Presently, the workflow involves extensive human effort in drafting, evaluating, and approving tender documents across three primary categories—Goods, Works, and Consultancy & Services. Key activities such as document preparation, content version control, Pre-Qualification Criteria (PQC) framing, bid evaluation, and award recommendation are largely manual, relying on expert memory, historical documents, and decentralized data.
	Name of the Project under which technology is developed	AI-based Procurement RFx Automation for NRL Refinery
	Funding Agency	Numaligarh Refinery Limited (NRL)
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	Currently, the tender process for Goods, Works, and Consultancy & Services is largely manual. Activities like document drafting, version control, PQC framing, bid evaluation, and award recommendation depend heavily on human effort, expert knowledge, and scattered historical data. This workflow introduces a semi-automated, AI-driven system using LLMs, NLP, and structured databases to assist in content creation, document comparison, PQC suggestions, and evaluation matrix generation. Integrated with existing platforms like SAP, it aims to streamline data

Sr. No.	Parameters	Brief details
		handling, reduce manual effort, enhance standardization, and improve procurement efficiency and compliance. Features • Generation of Technical Specification Document. • Preparation of extended procurement documents (Annexures with Pre-Qualification Criteria (PQC) sections). • Generation of Notice Inviting Tender (NIT) with Finalized Payment Terms. Technical bid evaluation report, Commercial bid evaluation report together with rejection/approval note.
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Ramesh Naidu Laveti, C-DAC Bangalore **ph:** +91 98443 34898, Mobile: +91 91641 73007 Email: **rameshl@cdac.in**
7.	**Brief details of Technology Developed**	Sightsaver is an AI based technology developed to screen retinal fundus images for detection and classification of the chronic illnesses like **Diabetic Retinopathy, Glaucoma, Hypertensive Retinopathy and Vascular Occlusion** from retinal fundus images at early stages.
	Name of the Project under which technology is developed	Design & Development of Artificial Intelligence framework using Data Analytics for detection of Chronic Illness from Retinal Images
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	6
	Brief Write-up on the Technology Developed	- The technology developed is able to classify Diabetic Retinopathy, Hypertensive Retinopathy, Vascular Occlusion, & Glaucoma. - The state-of-the-art AI models can detect and grade Diabetic Retinopathy into five different classes including Normal fundus images, Mild Non-Proliferative Diabetic Retinopathy, Moderate Proliferative Diabetic Retinopathy, Severe Non-Proliferative Diabetic Retinopathy and Proliferative Diabetic Retinopathy. - The deep learning-based models can accurately classify different grades of Hypertensive Retinopathy and transfer learning-based model can locate the Arterio-Venous nicking in the image. - Image Quality Assessment model is used to extract features based on texture to filter out images on the basis of quality, sharpness and entropy of the images to either accept or reject images based on quality. - Other state-of-the-art machine learn models are also developed for detection of Retinal Vein Occlusion, Glaucoma and Cataract with more than 95% accuracy. - The solution was validated during clinical trials at GMCH, Chandigarh. - Integrated with GenAI for automated report generation.
	Details of Contact Person along with Mobile No. and e-mail ID	Ms. Sonia Dosanjh, C-DAC Mohali soniad@cdac.in, +91 9876072229
8.	**Brief details of the technology developed**	An AI-powered computer vision-based production monitoring and counting system that automatically detects, counts, and tracks manufactured units in real time from live camera feeds. The system provides live production status, batch-wise analytics, alerts, and visual verification through a

Sr. No.	Parameters	Brief details
		centralized web dashboard, reducing manual intervention and improving production accuracy.
	Name of the Project under which the technology is developed	Industrial Automation System Using Vision AI
	Funding Agency	Internal funding
	Technology Readiness Level (TRL)	TRL-6
	Brief Write-up on the Technology Developed	It is an intelligent production monitoring solution designed for high-volume manufacturing environments. The system employs deep learning-based computer vision models to detect and count products moving on a production line in real time. It automatically tracks production batches against predefined targets, captures timestamped visual proofs, and generates alerts when targets are completed or anomalies are detected. The solution includes an AI processing backend, industrial camera integration, and a web-based dashboard for live monitoring, batch history, analytics, and event logging. It minimizes human error, improves production accuracy, enables real-time decision-making, and supports integration with existing ERP/MES systems. The technology is scalable and deployable in both on-premises and cloud environments.
	Details of the Contact Person along with the Mobile No. and e-mail ID	Ritesh Dhote, C-DAC Patna 9970364664 riteshd@cdac.in
9.	**Brief details of Technology Developed**	End-to-End Workflow Tool for Computer Vision
	Name of the Project under which technology is developed	Subsea Panoramic imaging System with AI Based Solution for Species Identification
	Funding Agency	Ministry of Earth Sciences
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	The **End-to-End Workflow Tool for Computer Vision** is designed to streamline the development, deployment, and monitoring of computer vision models. It **enables** users to upload data, perform annotation (both manual and automated), review annotations, train models, run inference, evaluate performance, and deploy models with minimal effort. By providing an integrated, all-in-one platform, the tool significantly accelerates the development and operationalization of vision AI applications. It caters to a broad range of users, including product teams, researchers, and domain experts across diverse sectors such as **healthcare, agriculture, manufacturing, smart cities, surveillance and security, retail, autonomous systems, remote sensing, and industrial inspection**, making it a versatile and comprehensive solution for end-to-end computer vision workflows.
	Details of Contact Person alongwith Mobile No. and e-mail ID	S Irene, C-DAC Chennai irenes@cdac.in 9940320384

Sr. No.	Parameters	Brief details
10.	**Brief details of Technology Developed**	Artificial Intelligence (AI) models are inherently vulnerable to adversarial samples which are maliciously crafted samples to make those models misbehave. The number of deployments of AI models in defense applications/products are increasing and are vulnerable to adversarial attacks. Therefore, the systematic evaluation of robustness of AI models under varied configurations of different attacks and defense capabilities is essential. Hence, it is essential to dynamically adopt a defense method to tackle an attack w.r.t the application requirements. These critical issues are largely unsolved due to lack of a comprehensive framework to evaluate the robustness of AI models. With this motivation, we developed a framework titled **"Dynamic Adversarial Robustness Analysis Framework for AI Applications (DARFA)"** for analyzing the robustness of AI models and in-turn the AI applications.
	Name of the Project under which technology is developed	AI for Cyber Security (AICS) under National Super Computing Mission (NSM)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	Current TRL 5
	Brief Write-up on the Technology Developed	DARFA aims to provide an easy-to-use framework that enables the users to comprehensively evaluate the robustness of AI models, providing insightful decisions to take the necessary actions against adversarial attacks thereby strengthening the monitoring and reviewing of AI applications to make them trustworthy. Features • Zero-Coding - The ability to conduct experiments using a user-friendly GUI, eliminating the need for writing code. • State-of-the-art Attack & Defence Algorithms - Capacity to accommodate a wide range of attack and defence methods, along with the provision of appropriate criteria for assessing the effectiveness of these methods. • Comprehensive Robustness Evaluation Report Generation - Detailed report of robustness analysis of the user AI model after simulation of adversarial attacks and defences are generated. • Supporting multiple data modalities (Tabular, Images, Text, Audio/Speech, Video etc.) • HPC-Enabled - The integration of HPC capabilities to enhance the efficiency and scalability of robustness evaluation processes. Deployment Model - Can offer as a service on NSM HCP clusters, Stand-alone deployment on servers and Labkit for student community
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